

Inference_1_training_overlap

2026.09.03 - The nominations are novel at the query-double level and enriched at the pair level

Follows the ensembled review in `[[experiments.010-kuzmin-tmi.scripts.review_inference_1_top_triples]]`, which ranked **inference_1** by the mean of three checkpoints but reported only the negative tail and said nothing about how much of each nominated triple the screen had already measured.

Script: `experiments/010-kuzmin-tmi/scripts/inference_1_training_overlap.py`.

The screen has two roles, and the 526 genes sit mostly in one

Kuzmin crosses a query double mutant against an array of single mutants, so a training record is a query pair plus one array gene. On this build that reading is exact: 420 pairs occur five or more times and cover 376,733 pair-instances against 376,732 records. Those 420 doubles involve 835 query genes; 4,263 genes appear as array genes.

role of the 526 inference_1 genes	genes	share
query gene, member of a query double	116	22%
array gene only	333	63%
never perturbed in the build	77	15%

Query-double overlap is essentially zero, and structurally so

Only 2,430 of 4,370,595 triples carry one of the 420 query doubles, 0.056 percent. Of the top 200 positive predictions, zero do. Of the 100,000 most negative, one does.

That is not the ranking avoiding them. Only **22 of the 420 query doubles have both members inside the 526-gene space at all**, so almost no inference triple could carry one. The gene selection for **inference_1** was built from morphology, expression and metabolic criteria, not from the trigenic screen’s query list, and the two barely intersect.

The consequence is the important part. **inference_1** is very nearly a query-pair-disjoint space by construction, which is the regime where the additive null falls from 0.400 to 0.127 plus or minus 0.033 and the nonlinear baseline drops below it. Every nomination here lives in that regime, and the transformer has not been measured in it.

Pair overlap is low overall but rises steeply in both tails

Counting how many of a triple’s three gene pairs ever co-occur in any training record, recurring or not:

pairs already screened, of 3	triples	share
0	3,153,484	72.2%
1	816,705	18.7%
2	319,381	7.3%
3	81,025	1.9%

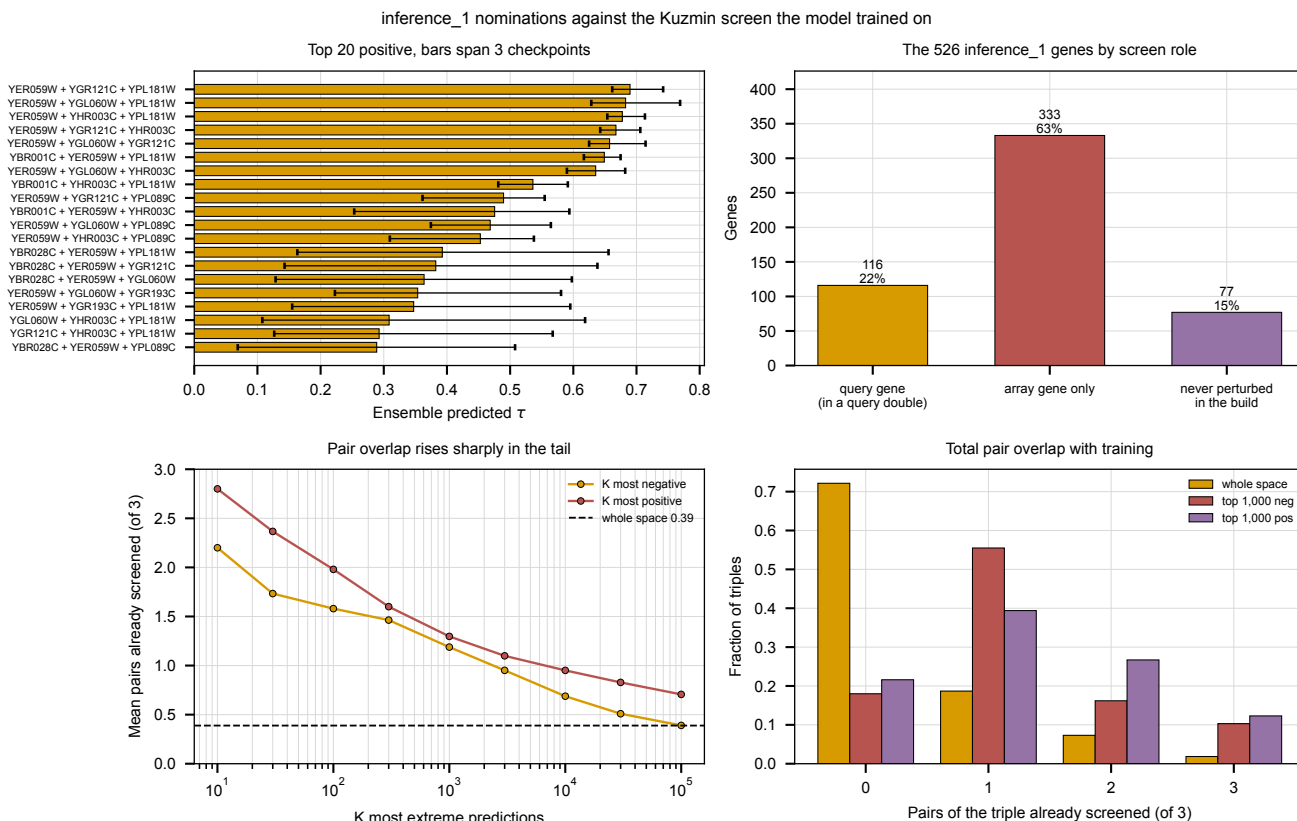
The whole-space mean is 0.389 of 3. In the tails it is far higher: 2.20 for the ten most negative and 2.80 for the ten most positive, a 5.7 and 7.2 times enrichment, decaying smoothly back to the baseline by $K = 100,000$.

So the extreme predictions are not indiscriminate extrapolations. They concentrate on triples whose constituent pairs the screen actually co-measured, even though the query double itself is novel. **Hypothesis (untested):** the model is composing measured pairwise structure rather than recalling a screen, which would be the favorable reading. Distinguishing that from a pair-frequency artifact needs the disjoint-split retrain, since a model that scores partly on pair familiarity would produce this same enrichment.

The highest predictions

Top 200 positive run +0.690 down to +0.082, against a training-label standard deviation of 0.0633. They are more concentrated than the negative tail: 65 distinct genes across the top 200, with YER059W in 123 of them. The negative tail's comparable figures are 113 distinct genes and YHLO29C in 110.

Across-checkpoint spreads on the leaders are 0.024 to 0.062, tighter in relative terms than the negatives, whose median spread is 0.0694. Positives remain the direction with no retrieval validation, since the published Kuzmin call is one-sided negative.



Outputs

- experiments/010-kuzmin-tmi/results/inference_1_overlap_summary.json
- experiments/010-kuzmin-tmi/results/inference_1_overlap_by_k.csv
- experiments/010-kuzmin-tmi/results/inference_1_top_positive.csv